

MOUNT EDGECOMBE, South Africa

WMO: 685830

Lat: 29.7067S

Lon: 31.0461E

Elev: 94

StdP: 100.20

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	10.7	1.3	4.2	15.6	3.2	4.8	15.1	6.8	19.1	6.1	19.5	0.7	260	0.261

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.4	30.4	23.6	29.3	23.2	28.3	22.8	24.8	28.6	24.2	27.9	23.7	27.2	3.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.7	27.2	23.0	18.0	26.8	22.5	17.4	26.3	76.0	28.8	73.6	28.0	71.6	27.4	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.9	5.2	7.9	33.9	1.7	1.6	6.6	35.1	5.6	36.1	4.7	37.0	3.4	38.2		
(5)			5.6	26.2	1.7	1.5	4.4	27.3	3.4	28.2	2.4	29.0	1.2	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.7	23.9	24.3	23.4	21.4	19.6	17.8	17.1	18.3	19.1	20.1	21.2	22.7		
	DBStd	3.17	1.98	1.87	2.16	2.09	2.16	1.87	1.88	2.07	2.41	2.26	2.26	2.27		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	151	0	0	1	2	12	31	47	26	19	9	3	1		
	CDD10.0	3906	430	399	416	343	297	233	220	256	272	312	336	393		
	CDD18.3	1016	172	166	158	96	49	14	8	24	41	63	89	136		
	CDH23.3	4740	1020	998	827	306	159	54	46	75	124	172	321	638		
(13)	CDH26.7	767	178	183	128	38	24	8	7	9	21	26	44	102		
(14)	Wind	WSAvg	2.0	2.2	2.0	1.9	1.7	1.5	1.6	1.9	2.2	2.4	2.4	2.3		
(15) Precipitation	PrecAvg	858	120	89	87	50	32	20	44	34	58	98	100	125		
	PrecMax	1153	301	228	205	137	144	69	193	110	200	273	263	320		
	PrecMin	422	48	15	29	9	0	0	3	12	21	32	41			
	PrecStd	198	57	58	52	34	35	20	61	26	39	54	51	74		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	31.4	30.7	29.8	29.4	27.7	27.5	27.7	29.2	29.8	30.3	31.2		
		MCWB	24.5	25.1	24.4	22.2	19.7	17.1	16.2	18.2	19.9	22.3	23.6	24.9		
	2%	DB	29.6	29.8	29.2	27.5	26.5	24.6	24.4	25.2	26.4	26.9	27.8	28.8		
		MCWB	23.9	24.1	23.6	21.8	19.8	16.5	16.4	17.7	19.8	20.7	22.2	23.4		
	5%	DB	28.5	28.7	28.1	26.2	25.0	23.0	22.5	23.6	24.6	25.1	26.4	27.6		
		MCWB	23.4	23.6	23.2	21.3	19.1	16.7	16.1	17.5	19.1	20.1	21.6	22.8		
	10%	DB	27.5	27.7	27.0	25.0	23.6	21.6	21.1	22.3	23.1	23.8	25.1	26.5		
		MCWB	22.9	23.0	22.7	20.7	18.7	16.2	15.6	17.0	18.4	19.6	21.0	22.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.8	25.1	23.4	22.2	19.5	18.8	20.2	21.6	22.6	24.2	25.1		
		MCDB	29.5	30.1	29.4	27.9	26.7	23.6	23.5	25.2	26.3	28.1	28.7	29.7		
	2%	WB	24.5	24.8	24.3	22.6	20.9	18.4	17.8	19.1	20.4	21.5	23.0	24.1		
		MCDB	28.4	28.3	28.0	26.4	24.9	22.4	22.1	23.2	24.7	25.4	26.6	27.6		
	5%	WB	23.9	24.2	23.7	21.8	20.0	17.6	17.1	18.4	19.7	20.8	22.2	23.4		
		MCDB	27.6	27.7	27.1	25.1	23.6	21.6	21.1	22.2	23.4	24.1	25.4	26.7		
	10%	WB	23.3	23.6	23.1	21.2	19.2	16.8	16.3	17.7	19.0	20.1	21.5	22.7		
		MCDB	26.9	27.0	26.4	24.3	22.6	20.6	20.0	21.4	22.4	23.2	24.6	25.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.4	6.4	7.1	8.7	9.8	9.7	8.5	7.1	6.3	6.2	6.1		
		MCDBR	8.3	8.0	7.8	8.9	10.6	11.7	12.1	10.9	10.3	9.2	8.7	8.2		
		MCWBR	4.1	3.9	3.9	4.4	5.5	6.4	6.6	5.6	5.4	5.1	4.9	4.3		
	5% WB	MCDBR	7.7	7.4	7.4	8.0	9.1	10.1	10.6	9.0	8.9	7.9	7.6	7.5		
		MCWBR	4.0	3.8	4.0	4.2	5.2	6.2	6.5	5.1	5.0	4.7	4.5	4.2		
(40) Clear-Sky Solar Irradiance	taub		0.453	0.442	0.435	0.404	0.378	0.362	0.372	0.435	0.485	0.497	0.468	0.461		
	taud		2.303	2.338	2.340	2.386	2.424	2.445	2.404	2.216	2.099	2.110	2.225	2.264		
	Ebn at Noon		893	884	856	830	804	795	801	785	795	825	873	888		
	Edh at Noon		140	131	124	107	93	86	93	125	155	163	150	146		
(44) All-Sky Solar Radiation	RadAvg		5.92	5.81	5.06	4.12	3.52	3.14	3.35	3.90	4.50	4.91	5.41	5.84		
	RadStd		0.47	0.43	0.25	0.25	0.18	0.15	0.22	0.23	0.29	0.44	0.46	0.43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page